Quantifying the Cardiovascular Response to Varying Degrees of Hemorrhage for the Establishment of a CFD Model

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Introduction: Hemorrhage, characterized by the loss of blood, remains a significant concern in both civilian and military settings, contributing to approximately 30-40% of trauma mortality worldwide.[1] Understanding the cardiovascular response to varying degrees of hemorrhage is crucial for improving patient outcomes and developing effective lifesaving interventions. In this study, we aimed to quantify the cardiovascular response to different levels of hemorrhage in an established swine model of hemorrhagic shock. By analyzing changes in vessel diameter, cardiac output (CO), and pressure-volume (P-V) loops at 10%, 20%, and 30% hemorrhage levels, we sought to provide valuable insights into the underlying hemodynamic alterations associated with hemorrhage. Moreover, we aimed to utilize these findings to build and refine a computational fluid dynamic (CFD) model for hemorrhage. Using the Cardiovascular Integrated Modeling and Simulation (CRIMSON) platform, we report preliminary findings that shed light on the hemodynamic changes associated with varying degrees of hemorrhage.

Materials and Methods: Yorkshire swine underwent controlled hemorrhage (10%, 20%, and 30% of total blood volume) over 30 minutes. Analysis focused on two key time points: T0 (baseline) and T30 (end of hemorrhage). Pigs (n=3) were instrumented with flow and pressure probes to acquire continuous hemodynamic data, and a P-V loop catheter was positioned in the left ventricle via apical stick to collect raw P-V data. Python algorithms were developed to process raw hemodynamic data to generate boundary conditions for the CFD model and quantify CO and stroke volumes (SV). To determine vascular diameter changes during hemorrhage, contrast-enhanced CT scans were performed on an additional 7 pigs. Using TeraRecon's AQi software (v.4.7, Durham, NC), vascular diameters at 34 specific locations were quantified, including the aorta, carotid, brachiocephalic, celiac, and renal arteries. These CT scans were then imported into CRIMSON® (v.1.4.5, Ann Arbor, MI) to create 3D geometric models of the aorta and its immediate branch vessels. The ascending aorta served as the inlet to the CFD model with the distal aorta and takeoff vessels serving as the outlets. The ascending aortic flow rate from the *in vivo* models served as the input boundary condition, while the outlet boundaries were connected to 0D three-element Windkessel models. The simulations assumed steady, incompressible flow with no-slip conditions at the arterial walls, and the arterial walls were assumed to be rigid. Models were calibrated to pressure and flow profiles obtained from the *in vivo* porcine hemorrhage models.

Results and Discussions: The ensemble-averaged P-V loops show that at baseline, the average SV and CO across all animals was 48.23 ± 31.15 mL and 5.65 ± 2.63 L/min, respectively. At T30, SV decreased to 45.8, 12.0, and 8.03 mL for 10%, 20%, and 30% hemorrhage, respectively (Figure 1A), which is at least half of the starting SV. Similarly, CO decreased to 5.18, 1.66, and 1.17 L/min for 10%, 20%, and 30% hemorrhage, respectively (Figure 1B). In our imaging cohort, we observed no significant changes ($p \leq .05$) in any of the vessel diameters between baseline (i.e., T0), 10%, 20%, and 30% hemorrhage, indicating a consistent maintenance of large arterial vascular diameters with minor variations (Figures 1C-D). Simulations of 10% hemorrhage showed mean blood pressures of 55 mmHg across most of the aorta, while 20% hemorrhage had mean pressure of approximately 32 mmHg (Figure 1E). Additionally, at T30, 20% hemorrhage exhibited substantial reductions in mean blood velocity (cm/s) across all vessels compared to 10% hemorrhage, indicating a more severe impact on blood circulation (Figure 1E). These CFD simulations match well with our experimental data.

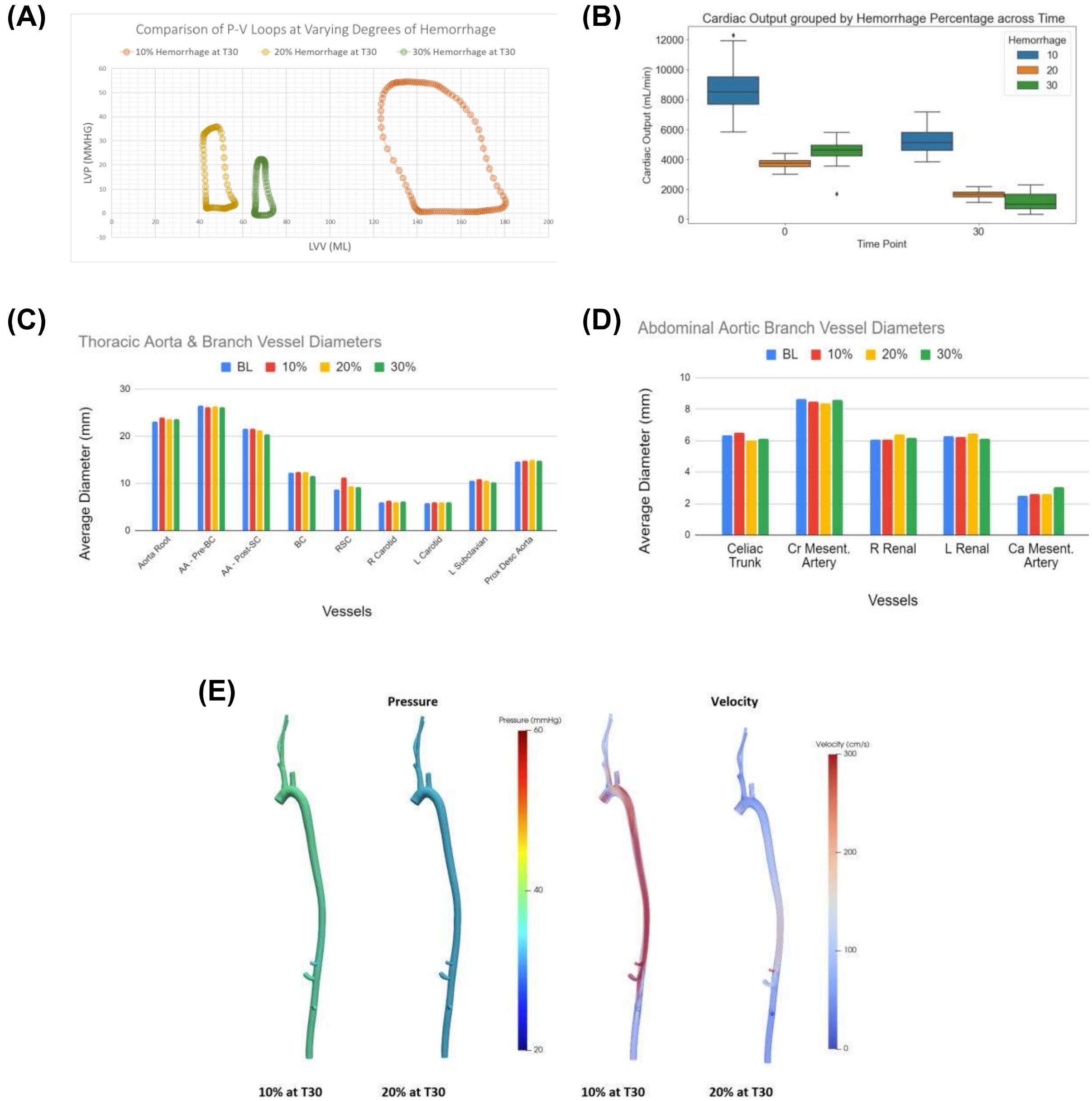

Figure 1: Quantification of Cardiovascular Response to Varying Degrees of Hemorrhage, **(A)** Ensemble Averaged P-V Loops for 10%, 20%, and 30% Hemorrhage at T30, **(B)** Cardiac Output for 10%, 20%, and 30% Hemorrhage at T0 and T30, **(C)** Average Vascular Diameter (mm) for Thoracic Aorta & Branch Vessels, **(D)** Average Vascular Diameter (mm) for Abdominal Aortic Branch Vessels, **(E)** Comparison of Pressure and Flow Between 10% and 20% Hemorrhage at T30 within CFD Model

Conclusion: In this study, the aorta and its greater vasculature displayed minimal diameter changes in response to varying degrees of hemorrhage up to 30% total blood volume, suggesting that most changes in vascular diameters during hemorrhage are likely to occur in smaller arterioles and/or the microvasculature. Furthermore, we observed substantial reductions in SV and CO, even at low levels of hemorrhage (e.g., 10%). The 40% decrease in SV and average CO at 10% hemorrhage indicates that there may be a nonlinear relationship between total blood volume lost and cardiac performance. These observations have important implications for our understanding of vascular response to hemorrhagic conditions and how to improve the fidelity of CFD models. Nevertheless, these results indicate a direct relationship between the severity of hemorrhage and its impact on the circulatory system. Further studies are needed to better understand the underlying mechanisms regulating vascular behavior during hemorrhagic events, and identifying the critical thresholds for cardiac parameters that can contribute to the advancement of diagnostic and therapeutic approaches for critically injured patients.

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